

Sathya Devarajan

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Purdue University — John Martinson Honors College

West Lafayette, IN

B.S. Mechanical Engineering, GPA: 3.8/4.0

May 2028

Experience

Lockheed Martin Sikorsky

Stratford, CT

Tooling Design Engineer Intern

June 2026 – August 2026

- Designed & revised 14 production tooling fixtures & jigs (drill jigs, assembly locators, masking fixtures) in CATIA V5 for CH-53K, Black Hawk, & search-and-rescue rotorcraft; 3 now in use on shop floor; 11 in procurement & manufacturing
- Toleranced 12 tools to ASME Y14.5 GD&T, defining datum reference frames and profile/position callouts (± 0.001 in. to ± 0.015 in.) matched to CNC milling, manual machining, and additive processes
- Traced kinematic underconstraint and lack of sealing in 3 released production tools back to root cause; redesigned geometry to eliminate secondary mill setups and reduce vendor procurement lead time by 2 weeks
- Maintained multi-level BOMs for all 14 tools & routed engineering change orders through SAP ERP & 3DEXPERIENCE PLM, so manufacturing engineers & machinists worked from current tooling data across releases
- Led mechanical design on hybrid propulsion system intern project for rotor-blown Nomad VTOL; packaged ICE, motor generator, battery pack, & a dual motor & planetary gearbox 20% under 80 lb weight limits; presented to leadership

Purdue Electric Racing (FSAE Electric)

West Lafayette, IN

Rear Upright Owner

June 2026 – Present

- Designed rear upright & planet carrier for 2027 outrunner drivetrain in NX, cutting weight 32% (.44 lb shaved per corner) while maintaining rigid suspension connections & aligning carrier geometry with front uprights for manufacturing ease
- Executed nonlinear static structural FEA on the rear upright & hub group targeting SF 1.5; optimized rib topology & fillet radii against localized stress concentrations to reduce suspension bracket compliance from 0.079 in. to under 0.02 in.
- Resolved stress propagation issues in Ansys via strain-energy study & first principles calculation; identified bonded bearing contacts on radial and abutment interfaces as problematic & created shared topology in Ansys discovery & switched to frictionless & no-separation contacts on specific interfaces for correct stress propagation & deformation
- Ran mesh convergence to produce highest fidelity results on max cornering, accel, & combined tire contact patch cases to verify previous studies & geometry modifications; conducted fatigue study to ensure components met lifespan requirement

Battery Thermal Engineer

June 2026 – Present

- Enhanced lumped-parameter battery thermal model accuracy by calculating effective thermal conductivities from empirical literature & test datasets, replacing uniform isotropic cell assumptions to increase thermal simulation fidelity

Drivetrain Engineer

August 2025 – May 2026

- Detailed 2D production drawings for rear upright assemblies, applying ISO P7 interference limits & ISO H8 clearance fit on 3.937 in. & 4.528 in. nominal bearing seats (-0.0009 in. / -0.0022 in. & $+0.002$ in. / 0 in.) to maintain a press fit for SKF 61818 bearings across operating thermal ranges in 7075-T6 aluminum
- Cut wheel-upright machining time 50% by reworking the toolpaths from 3-axis to 5-axis CNC milling, then built the fixture plate that locates and holds the upright through the remaining 3-axis production runs
- Single-point bored 3.937 in. & 4.528 in. bearing seats on a manual vertical knee mill using an adjustable micrometer boring head; verified bore roundness and diameter within 0.001 in. via telescoping gauge & micrometer
- Turned 8620 alloy steel gear blanks on manual lathes prior to external gear wire EDM, holding minimum outside diameters and a face width tolerance of ± 0.004 in. to meet required gear contact area & expected lifespan

Projects

Custom Electric Bike Design

October 2025 – Present

- Built a fully parametric frame assembly in Siemens NX keyed to off-the-shelf tubing dimensions, so wheelbase, head angle, chainstay angle, and dropout geometry all update from a single set of driving inputs
- Executed Ansys static structural FEA on each candidate frame, swapping stock tube diameters and wall thicknesses through the parametric model to find the lightest combination that holds a 2.0 minimum safety factor under static load
- Ran cost-versus-performance trade studies on motors, sprockets, and tubing alongside the FEA, converging on the cheapest catalog set that still meets the target safety factor and ride geometry, 9% under the first build cost

Skills

CAD & PLM: CATIA V5, Siemens NX, Fusion 360, 3DEXPERIENCE, Teamcenter, SAP ERP

Manufacturing & Metrology: GD&T (ASME Y14.5), 3/5-Axis CNC CAM Programming, Manual Lathe/Mill, Rapid Prototyping (FDM), Fixture Design, Soldering

Tolerancing & Analysis: Ansys Mechanical (Static Structural), Dive CFD

Programming & Instrumentation: Python (NumPy, Matplotlib), MATLAB, Data Acquisition, LTSpice, Arduino